2 The Pilot

The National Strategy for Financial Education (ENEF) was established by the Brazilian Federal Government in 2010 to promote financial education and empower individuals to make autonomous financial decisions.[3] In 2015, the Brazilian Association of Financial Education (AEF-Brasil),[4] ENEF's implementing partner, developed a financial educational pilot targeting elementary and middle school students, for which the municipalities of Joinville and Manaus volunteered to participate. The intervention was delivered throughout the 2015 school year in a sample of public elementary and middle schools (grades 1–9) managed by the respective municipal governments. The initiative followed an earlier financial literacy program implemented under the same strategy in 892 public high schools across six Brazilian states, which yielded significant improvements in students' financial knowledge, intention to save, and financial autonomy, as well as greater participation in their household finances (Bruhn et al., 2016).

The two pilot municipalities have distinct socioeconomic contexts. Manaus, the capital of Amazonas, is in the less affluent North, while Joinville, in Santa Catarina, is in the relatively affluent South. At the time of the pilot, Joinville exhibited higher values on development indicators, including GDP per capita (USD 23,500 vs. USD 16,300, in 2015 PPP terms), the Municipal Human Development Index (0.809 vs. 0.737), and school attendance among 6- to 14-year-olds (97% vs. 94%).[5] Together, the two settings span two of the major socioeconomic profiles found in the Brazilian education system, broadening the external validity of our findings within the country while not exhausting its full diversity.

2.1 Teaching Materials

ENEF's teaching materials comprise one student book and one teacher book for each of grades 1–9, of which the pilot evaluates a subset, described below.[6] These materials follow the OECD's definition of financial education as the process by which individuals improve their understanding of financial concepts and risks and develop the skills and confidence to make informed financial decisions (OECD, 2005). This process operates through three means: information (knowledge

[3] ENEF stands for Estratégia Nacional de Educação Financeira and was created by a Federal Decree – 7.397 of December 22, 2010. It brings together the Ministry of Education, the Central Bank of Brazil, the Securities and Exchange Commission of Brazil (CVM), the Ministry of Finance, the Superintendence of Private Insurance (SUSEP), and the Superintendence of Complementary Pension (PREVIC).

[4] AEF-Brasil stands for *Associação de Educação Financeira do Brasil*.

[5] GDP per capita from the 2015 Municipal Accounts of the Brazilian Institute of Geography and Statistics (IBGE): R$ 47,233 for Joinville and R$ 32,598 for Manaus, converted to international dollars using the 2015 PPP conversion factor for Brazil of R$ 2.01 per international dollar (World Bank). Municipal Human Development Index and school attendance rates from the 2010 Demographic Census, the latest available at the time of the pilot, disclosed by the IBGE.

[6] Materials are publicly available on the CVM's investor education portal at `https://gmw.investidor.gov.br/ensino-fundamental/`. They were published in 2014 by the National Committee of Financial Education (CONEF) and validated by an inter-institutional pedagogical panel.

of concepts), instruction (attitudes and skills), and objective advice on specific financial products, of which the materials cover the first two, with greater emphasis on instruction across all grades.

The content is organized along two dimensions: a spatial dimension that moves from the individual to local, regional, national, and global levels, emphasizing how individual choices reverberate at larger scales; and a temporal dimension that connects past, present, and future decisions, placing the intertemporal nature of financial choice at the center. These dimensions are operationalized through four pedagogical modules, each targeting a grade range: a project-based approach for grades 1–4, a choose-your-own-adventure narrative for grades 5–6, a role-playing simulation game for grades 7–8, and a news-magazine format for grade 9.[7] Table 1 shows examples for the four pilot-grade books.

The content was not designed to be delivered as a stand-alone subject. Instead, it was designed to be integrated across subjects of the existing curriculum and to be taught by teachers of any specialty, drawing on what the teacher books call the teacher's role as a citizen rather than as a subject specialist. This integration is built into the material itself: the margins of the teacher books contain icons flagging, for example, connections to specific curricular areas (Portuguese, Mathematics, Environmental Education); the introduction of formal financial concepts (e.g., compound interest, household budget); or psychological biases (e.g., loss aversion, immediacy). We discuss in section 4 how this cross-curricular integration played out in the classroom.

Because each grade range receives a distinct combination of content and pedagogical format, our estimates capture the joint effect of these elements. We discuss grade-specific results with this combined nature in mind in section 6.

2.2 Sampling and Randomization

We defined the sampling frame using the 2015 School Census. In Joinville, we included all 72 elementary and middle schools managed by the municipal government. In Manaus, we sampled 129 of the 302 elementary and middle schools using a two-stage procedure.[8] NAO ENTENDI O PROCEDIMENTO COM O IDEB DESCRITO NO FOOTNOTE. To account for differences in infrastructure and socioeconomic context across the two municipalities, we stratified the

[7]Each pilot-grade teacher book describes its module's pedagogical approach in Part I, Section 4. The curriculum has 10 learning competencies: debating rights and duties (C01); participating in socially and environmentally responsible financial decisions (C02); distinguishing wants from needs of consumption and saving within the family's life project (C03); reading and interpreting simple texts in the financial education domain (C04); critically reading advertising texts (C05); participating in financial decisions considering real needs (C06); acting as a multiplier (C07); elaborating a basic financial plan with assistance (C08); and caring for oneself, nature, and common goods, considering both the immediate (C09) and future (C10) consequences of present actions.

[8]We first excluded 53 schools located in riverside communities, following the guidelines of the Department of Education, due to access constraints. Of the remaining 302 schools offering at least one target grade, we sampled 36 of the 202 schools that only offered elementary grades based on the 2013 IDEB quartiles of their grade 5 performance; 28 of the 35 schools that only offered middle school grades based on the 2013 IDEB quartiles of their grade 9 performance; and all 65 schools that offered grades 1–9.

Table 1: Financial education materials by pedagogical module

Module (grade range)	Pedagogical format	Central concepts and attitudes
Project-based (grades 1–4)	Four projects organized around concrete objects of everyday life: a ball (Production and Consumption), the household's expenses (Organization), the school (Care), and Book Day festivities (Planning).	Price composition; environmentally responsible consumption; needs vs. wants; income and expenses; care for shared public and private spaces; short-term planning and estimation.
Choose-your-own-adventure (grades 5–6)	Three choose-your-own-adventure stories plus a closing group project, centered on environmental sustainability and the "five R's" (rethink, refuse, reduce, reuse, recycle).	Waste; cash vs. credit purchases; value vs. price; conscious consumption; short-term planning; risk; patrimony; sustainability. Psychological traps introduced: ostentation, immediacy, selective perception, over-confidence.
Role-playing (grades 7–8)	A six-session role-playing game in which student groups represent different economic agents, negotiate proposals, and manage a running budget; new rules are added at each session.	Personal budget (income − expenses = balance); investment and the risk-return trade-off; financing and interest; the financial system; patrimony protection; borrowing. Additional traps: framing, sunk costs, loss aversion.
News-magazine (grade 9)	A print magazine modeled on the visual style of a news website, with reports, interviews, columns, a three-episode web-series, a forum, and hands-on exercises.	Household budgeting; expense categorization; consumer traps; social entrepreneurship; the Consumer Defense Code; public space, public budget, and taxes. Attitudes emphasized: autonomy, risk perception, discipline, and spending control.

Notes: The table summarizes the four pedagogical modules of the program (covering grades 1–9). Each row describes one module based on the corresponding pilot-grade book we examined (grades 3, 5, 7, or 9).
Source: Student and teacher books for grades 1–9, available at https://gmw.investidor.gov.br/ensino-fundamental/.

randomization by municipality and by school type: schools offering only elementary grades (grades 1–5), schools offering only middle school grades (grades 6–9), and schools offering all elementary and middle school grades (grades 1–9), generating six strata in total (Table A.2). Within each stratum, we randomly assigned half of the schools to treatment and half to control, resulting in 101 treated and 100 control schools.

Budgetary constraints shaped several aspects of the design. First, the pilot could not include all elementary and middle grades; we selected grades 3, 5, 7, and 9 to generate evidence on each of the four pedagogical modules. Second, we did not conduct a baseline student survey and instead relied on rich school-level administrative data to perform pre-treatment balance tests (section 3). Third, within each school, the program was not delivered to every class of the pilot grades: at the beginning of the 2015 school year, each school principal, in both treatment and control schools, selected a single class at each pilot grade to be included in the survey sample. Across the 201 pilot schools, the median number of classes per pilot grade was 3,[9] so principals typically chose one of about three classes per grade.

This within-school selection of classes is a potential source of bias, although random assignment was preserved at the school level. Because randomization preceded principals' class selection,

[9] 2015 School Census.

control-school principals had no reason to select classes on characteristics that would correlate with the (counterfactual) intervention. As we show in section 3, observable socioeconomic characteristics, teacher characteristics, and pre-treatment performance are balanced between treatment and control schools, with one exception: students in grade 9 in control schools display higher pre-treatment reading scores. If higher reading correlates with stronger financial proficiency, this imbalance would bias our estimates downward. We discuss the robustness of our estimates to this concern in section 6.